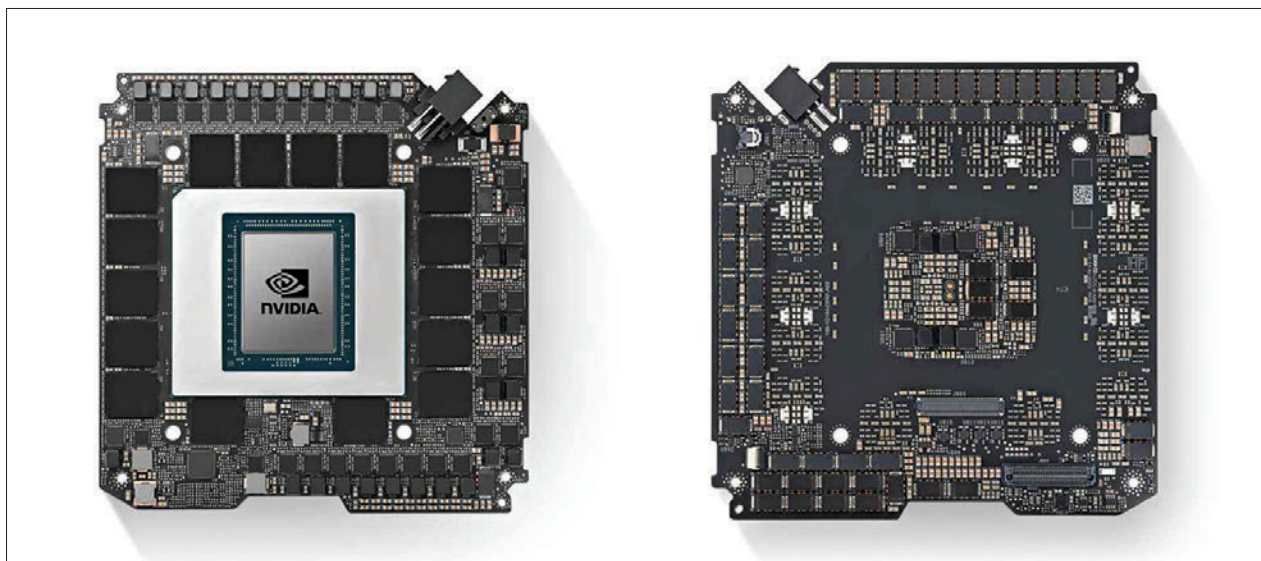




samples per minute in junkroom and 3.7K samples per minute in Classroom.

With Gaming performance, the RTX 5090 managed to hit 246 FPS at 1080p, 207 FPS at 1440p and 108 FPS at 4K. The RTX 4090 was about 231 FPS, 155 FPS and 74 FPS at the same resolutions. That's a pretty good 45 per cent bump in performance at 4K resolution but at 1080p, this difference drops to 6 per cent. In Assassin's Creed Mirage, we saw FPS values starting at 205 FPS for 1080p which dropped to around 130 when the resolution was scaled to 4K. That's about 15-41 per cent higher than the 4080 Super. But the 4080 Super wasn't the best card of the RTX 40 series so when you bring the RTX 4090 into the picture, the performance difference narrows down quite a bit. In Assassin's Creed Mirage, the difference is merely 4-17 per cent. The story is quite similar across a lot of the other titles but there are some games where the difference is around 25-33 per cent such as CyberPunk 2077.

POWER CONSUMPTION AND THERMALS

The NVIDIA GeForce RTX 5090, while a powerful graphics card, does exhibit a substantial power draw. In our testing, we observed a peak power consumption of 588 Watts. This is higher than both the RTX 4090 and the RTX 3090, which had Total Graphics Power (TGP) ratings of 450W and 350W, respectively.

The thermals of the RTX 5090 are within expectations for a high-performance graphics card. Under load, the card reached a maximum temperature of 90.88 degrees Celsius. This is comparable to the RTX 4090 and RTX 3090, both of which operated within a similar temperature range under load.

It's important to note that the power consumption and thermals of the RTX 5090 can vary depending on the specific workload and the efficiency of the cooling solution employed. While the card's power draw is higher than its predecessors, it's essential to consider the significant performance gains it offers.

AVAILABILITY AND PRICING

With a price tag of INR 2,14,000, the card is launched on January 30, 2025. However, potential buyers should be aware of several factors that could impact availability and pricing. Reports suggest that the RTX 5090 may be in short supply due to a combination of reasons. These could include manufacturing challenges, component shortages, or logistical issues. Additionally, the timing of the Chinese New Year, which falls close to the launch date, could temporarily disrupt production as factories close for the holiday period.

The high price point of the RTX 5090 is already a significant investment for most consumers. Coupled with the reports of limited availability, there are concerns

that opportunistic sellers may try to capitalise on the situation by further inflating prices. This could lead to price spikes in the initial weeks and months following the launch.

VERDICT - NVIDIA GEFORCE RTX 5090

The NVIDIA GeForce RTX 5090 is an aspirational product designed for gamers with deep pockets who demand the absolute best in graphics performance. At a price of INR 2,14,000, it is out of reach for most gamers in India.

The RTX 5090's DLSS with frame generation technology is impressive, delivering more frames with minimal latency. While the latency is noticeable, it doesn't make games unplayable.

However, in terms of raster performance, the improvement over the previous generation is not as significant. Given its price and limited accessibility, the RTX 5090 may not be the best option for value-conscious gamers.

—Mithun Mohandas

SPECIFICATIONS

CUDA Cores: 21760 | RT Cores: 318 | Tensor Cores: 680 | Base clock: 2017 MHz | Boost clock: 2407 MHz | VRAM: 32 GB | Memory interface: 512-bit | Process node: TSMC N4 | TDP: 575 W | Architecture: Blackwell

CONTACT

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